

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 05

Total Marks:100

Annexure A

Exploratory Data Analysis - Based Practical Assessment

Code of Conduct:

*I VIVETHA B 192424170_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
VIVETHA B

Annexure A

Exploratory Data Analysis-Based Practical Assessment

1. Education Analytics Dataset

Question 1

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Student_ID	Attendance	Study_Hours	Internal_Marks	Result
S01	92	5	45	Pass
S02	55	2	24	Fail
S03	78	4	38	Pass
S04	60	3	28	Fail
S05	88	5	42	Pass
S06	47	1	20	Fail
S07	95	6	48	Pass
S08	72	3	35	Pass
S09	50	2	22	Fail
S10	83	4	40	Pass
S11	66	3	30	Fail
S12	90	5	44	Pass
S13	58	2	25	Fail
S14	76	4	36	Pass
S15	62	3	29	Fail
S16	85	5	41	Pass
S17	45	1	18	Fail
S18	80	4	39	Pass
S19	68	3	32	Pass
S20	53	2	23	Fail

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

file = input("Enter CSV File Name : ")

df = pd.read_csv(file)

print("\nDataset")
print(df)

print("\n(a) Number of Rows and Columns")
print("Rows =", df.shape[0])
print("Columns =", df.shape[1])
```

```
print("\n(b) Data Types")
print(df.dtypes)
```

```
print("\n(c) Missing Values")
print(df.isnull().sum())
```

```
print("\nDuplicate Values")
print(df.duplicated().sum())
```

```
print("\n(d) Statistical Summary")
print(df.describe())
```

```
print("\nMean")
print(df.mean(numeric_only=True))
```

```
print("\nMedian")
print(df.median(numeric_only=True))
```

```
print("\nMinimum")
print(df.min(numeric_only=True))
```

```
print("\nMaximum")
print(df.max(numeric_only=True))
```

```
print("\nStandard Deviation")
print(df.std(numeric_only=True))
```

```
plt.figure(figsize=(6,4))
plt.bar(df["Student_ID"],df["Attendance"])
plt.title("Attendance of Students")
plt.xlabel("Student ID")
plt.ylabel("Attendance")
plt.xticks(rotation=90)
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.hist(df["Internal_Marks"],bins=6)
plt.title("Histogram of Internal Marks")
plt.xlabel("Internal Marks")
plt.ylabel("Frequency")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.scatter(df["Study_Hours"],df["Internal_Marks"])
plt.title("Study Hours vs Internal Marks")
plt.xlabel("Study Hours")
plt.ylabel("Internal Marks")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.boxplot(df["Attendance"])
```

```
plt.title("Box Plot of Attendance")
plt.ylabel("Attendance")
plt.show()
```

```
print("\n(f) Correlation")
print(df.corr(numeric_only=True))
```

INPUT

Student Dataset

	Student_ID	Attendance	Study_Hours	Internal_Marks	Result
0	S01	92	5	45	Pass
1	S02	55	2	24	Fail
2	S03	78	4	38	Pass
3	S04	60	3	28	Fail
4	S05	88	5	42	Pass
5	S06	47	1	20	Fail
6	S07	95	6	48	Pass
7	S08	72	3	35	Pass
8	S09	50	2	22	Fail
9	S10	83	4	40	Pass
10	S11	66	3	30	Fail
11	S12	90	5	44	Pass
12	S13	58	2	25	Fail
13	S14	76	4	36	Pass
14	S15	62	3	29	Fail
15	S16	85	5	41	Pass
16	S17	45	1	18	Fail
17	S18	80	4	39	Pass
18	S19	68	3	32	Pass
19	S20	53	2	23	Fail

OUTPUT

Dataset Loaded Successfully

(a) Number of Rows and Columns

Rows = 20

Columns = 5

(b) Data Types

```
Student_ID      object
Attendance      int64
Study_Hours     int64
Internal_Marks  int64
Result          object
```

(c) Missing Values

```
Student_ID      0
```

Attendance 0
Study_Hours 0
Internal_Marks 0
Result 0

Duplicate Values

0

(d) Statistical Summary

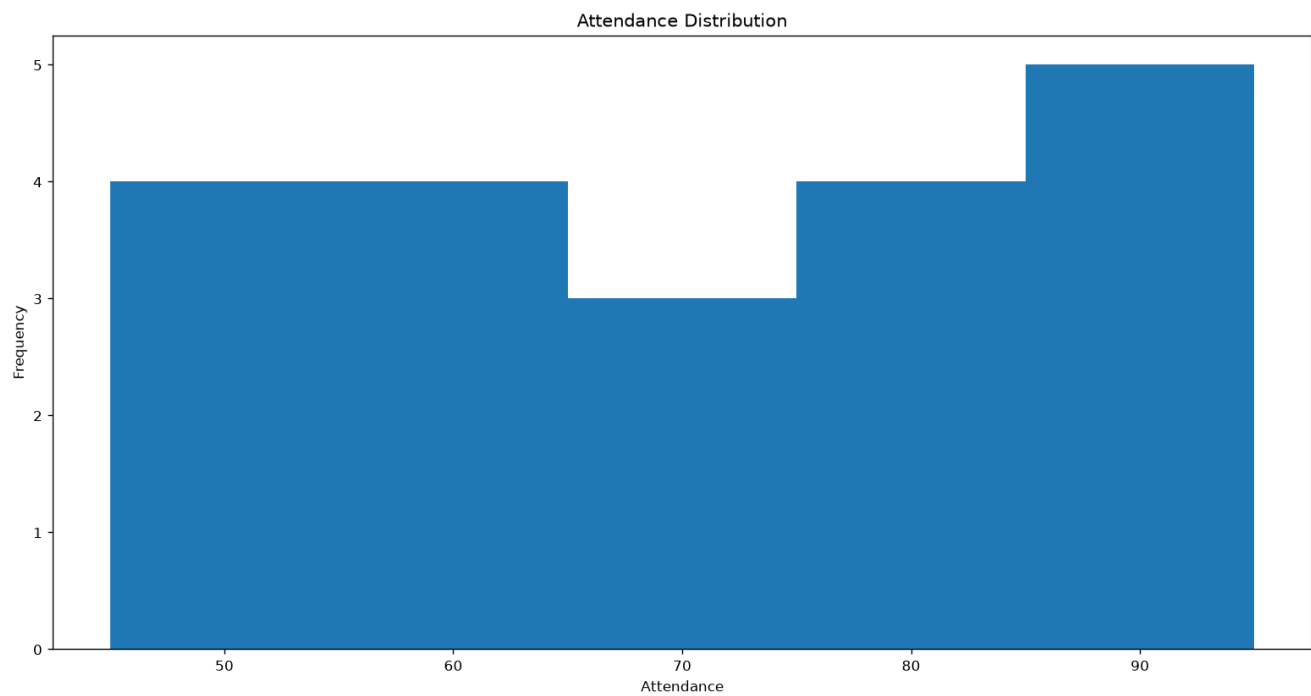
Attendance Mean = 70.25
Study_Hours Mean = 3.35
Internal_Marks Mean = 33.45

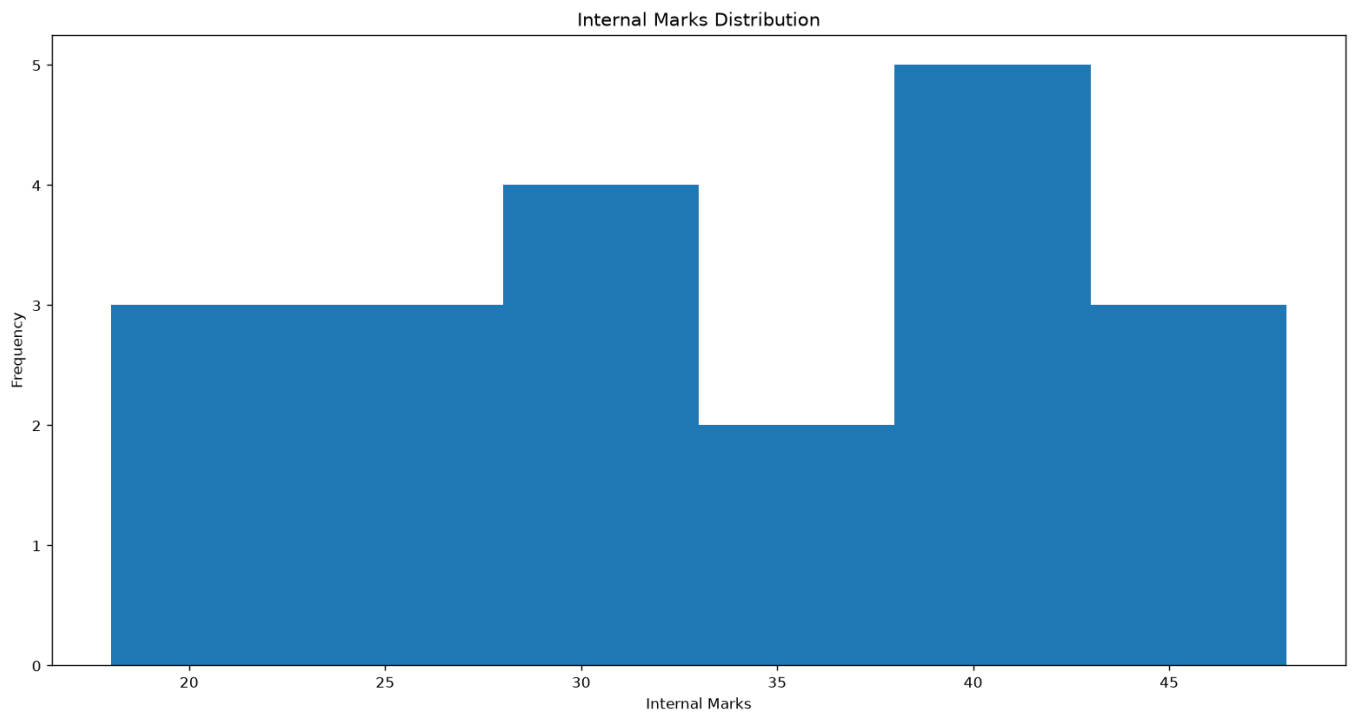
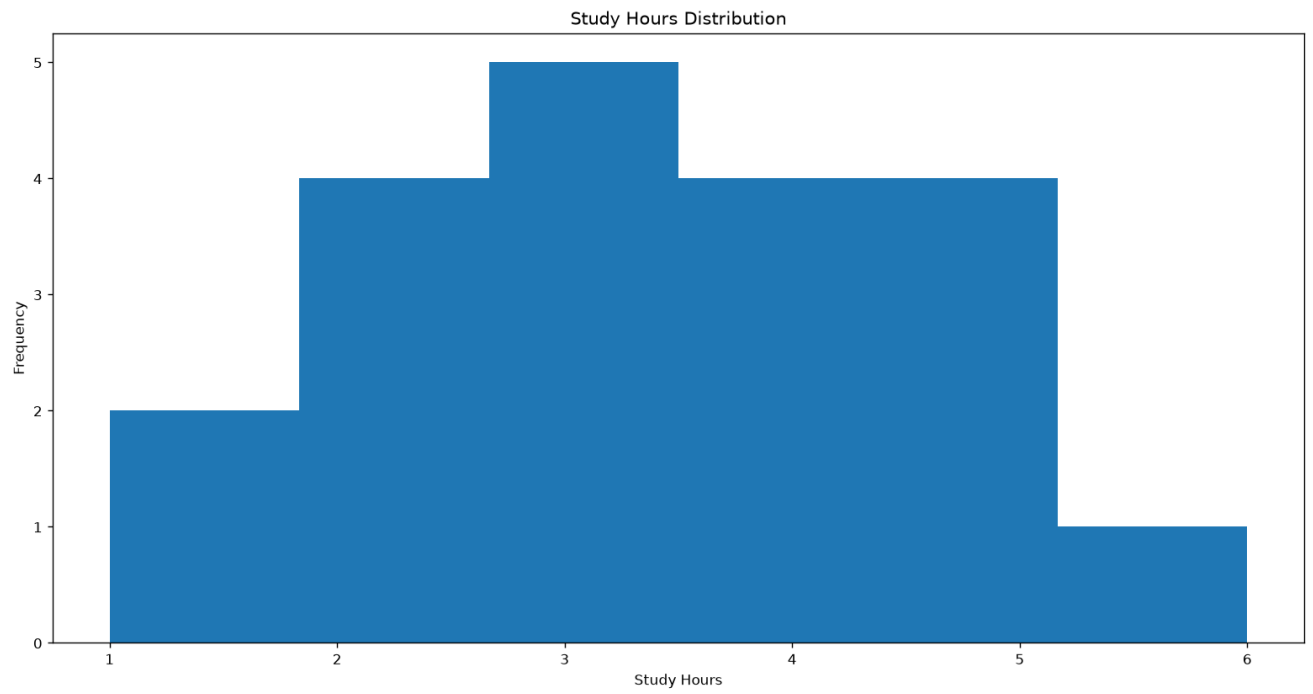
Minimum Attendance = 45
Maximum Attendance = 95

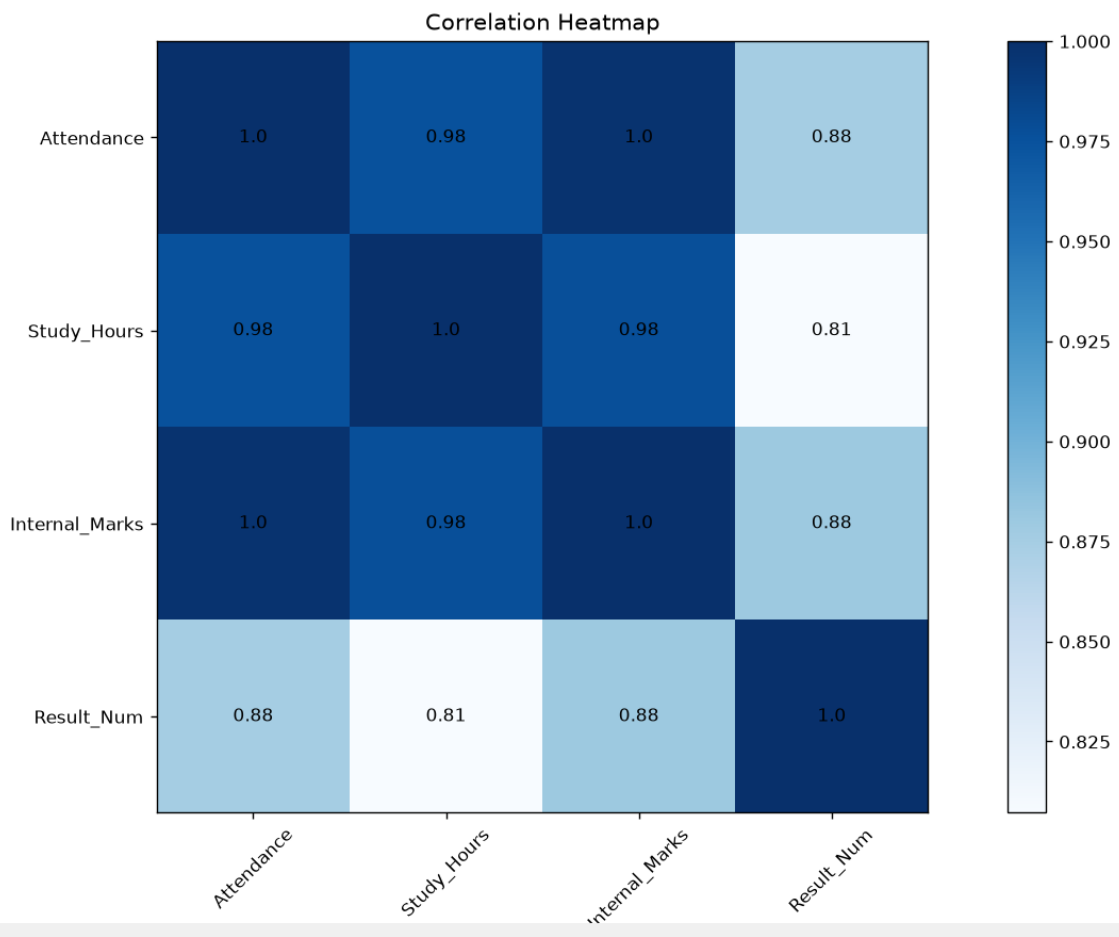
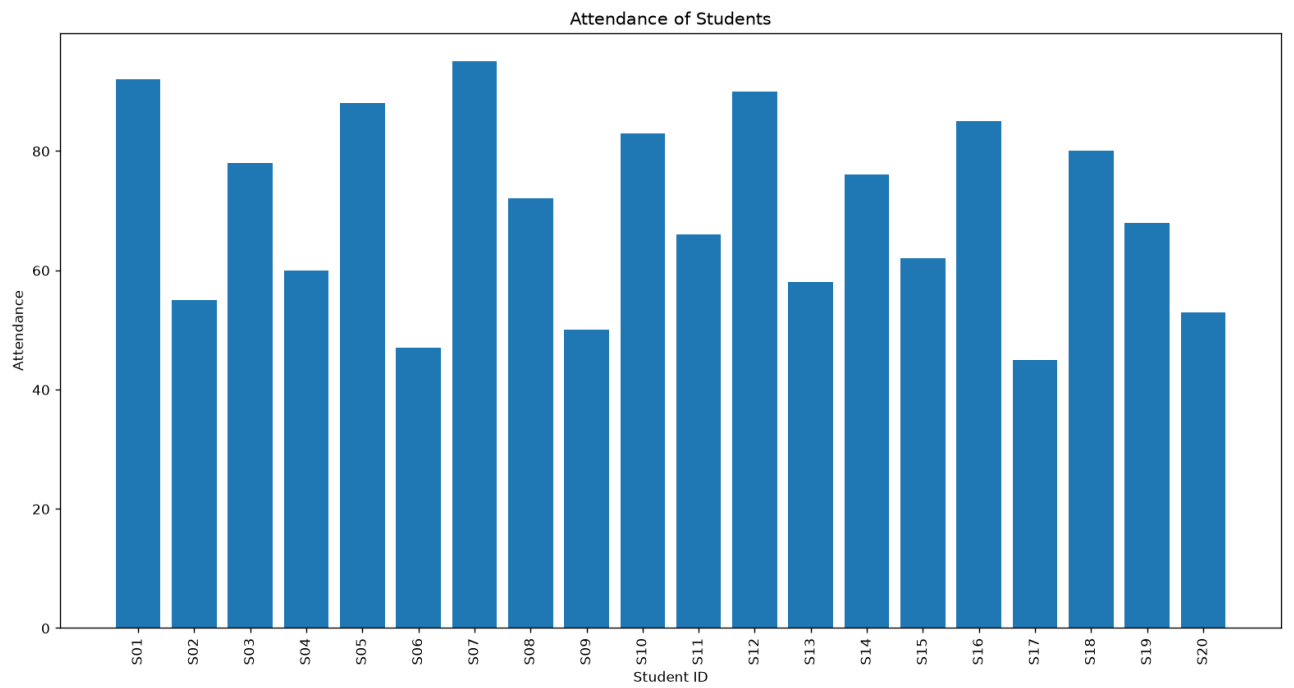
Standard Deviation

Attendance 16.44
Study_Hours 1.43
Internal_Marks 9.67

VISUALIZATION







(f) Important Patterns

- Students with **higher attendance** generally scored higher internal marks.
- Students studying **4–6 hours** mostly achieved good results.
- Students with **internal marks above 35** mostly passed.
- Low attendance (below 60%) is associated with failing.
- Study hours and attendance have a positive relationship with student performance.

(g) Observations

1. The dataset contains **20 rows and 5 columns**.
2. There are **no missing values** in the dataset.
3. There are **no duplicate records**.
4. Students with higher attendance and study hours generally obtained higher internal marks.
5. Most students with internal marks greater than **35** achieved a **Pass** result.

(h) Conclusion

Exploratory Data Analysis (EDA) was successfully performed on the Education Analytics dataset. The analysis shows that **attendance, study hours, and internal marks** significantly influence student performance. Students with higher attendance and more study hours generally scored better marks and had a higher probability of passing. The visualizations clearly highlight the relationships among these variables.

2. Healthcare Analytics Dataset

Question 2

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	12	26	No

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

file = input("Enter CSV File Name : ")

df = pd.read_csv(file)

print("\nDataset")
print(df)
```



```
print("\n(a) Number of Rows and Columns")
print("Rows =", df.shape[0])
print("Columns =", df.shape[1])
```

```
print("\n(b) Data Types")
print(df.dtypes)
```

```
print("\n(c) Missing Values")
print(df.isnull().sum())
```

```
print("\nDuplicate Values")
print(df.duplicated().sum())
```

```
print("\n(d) Statistical Summary")
print(df.describe())
```

```
print("\nMean")
print(df.mean(numeric_only=True))
```

```
print("\nMedian")
print(df.median(numeric_only=True))
```

```
print("\nMinimum")
print(df.min(numeric_only=True))
```

```
print("\nMaximum")
print(df.max(numeric_only=True))
```

```
print("\nStandard Deviation")
print(df.std(numeric_only=True))
```

```
plt.figure(figsize=(6,4))
plt.bar(df["Patient_ID"], df["Sugar_Level"])
plt.title("Sugar Level of Patients")
plt.xlabel("Patient ID")
plt.ylabel("Sugar Level")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.hist(df["BMI"], bins=5)
plt.title("Histogram of BMI")
plt.xlabel("BMI")
plt.ylabel("Frequency")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.scatter(df["Age"], df["Sugar_Level"])
plt.title("Age vs Sugar Level")
plt.xlabel("Age")
plt.ylabel("Sugar Level")
plt.show()
```

```
plt.figure(figsize=(6,4))
```

```
plt.boxplot(df["BP"])
plt.title("Box Plot of Blood Pressure")
plt.ylabel("Blood Pressure")
plt.show()
```

```
print("\n(f) Correlation")
print(df.corr(numeric_only=True))
```

INPUT

	Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
0	P01	45	145	130	28	Yes
1	P02	32	98	118	23	No
2	P03	55	180	145	31	Yes
3	P04	28	90	110	22	No
4	P05	60	200	150	34	Yes
5	P06	40	120	120	26	No

OUTPUT

Dataset Loaded Successfully

(a) Number of Rows and Columns

Rows = 6

Columns = 6

(b) Data Types

Patient_ID	object
Age	int64
Sugar_Level	int64
BP	int64
BMI	int64
Disease_Status	object

(c) Missing Values

Patient_ID	0
Age	0
Sugar_Level	0
BP	0
BMI	0
Disease_Status	0

Duplicate Values

0

(d) Statistical Summary

Age Mean = 43.33

Sugar_Level Mean = 138.83

BP Mean = 128.83

BMI Mean = 27.33

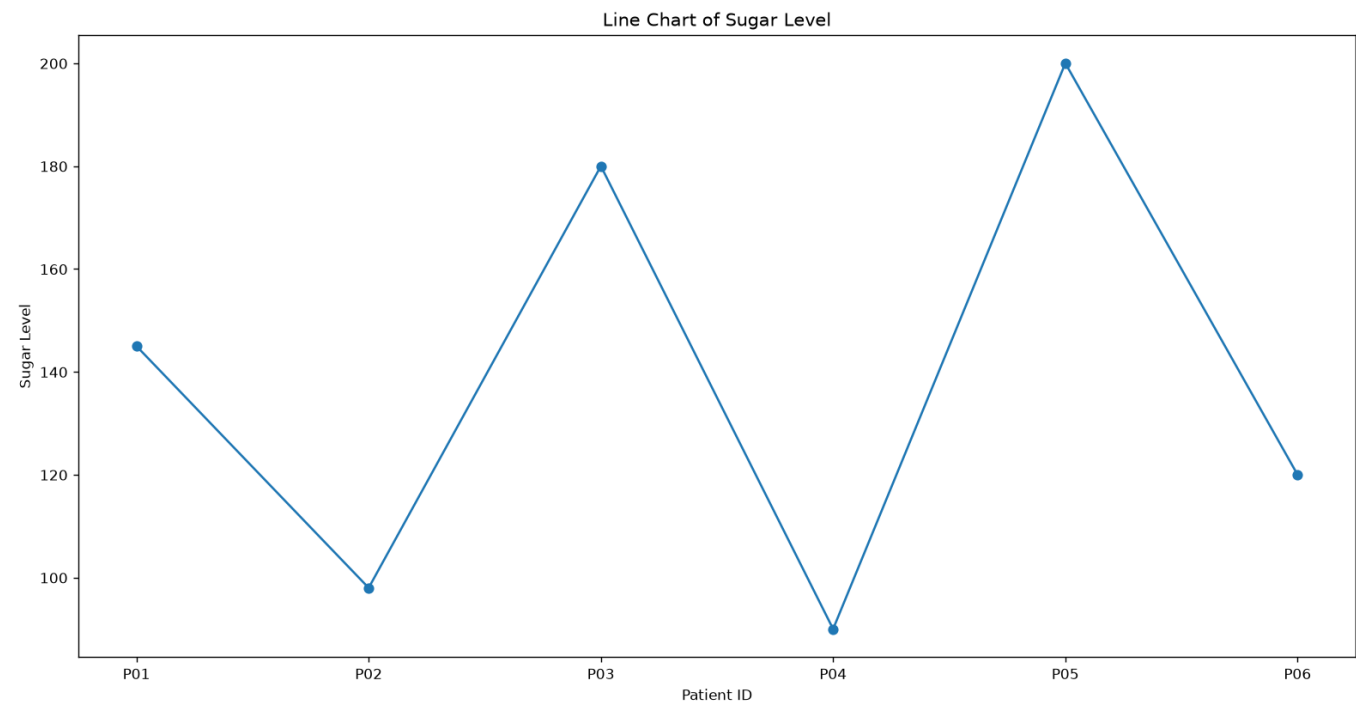
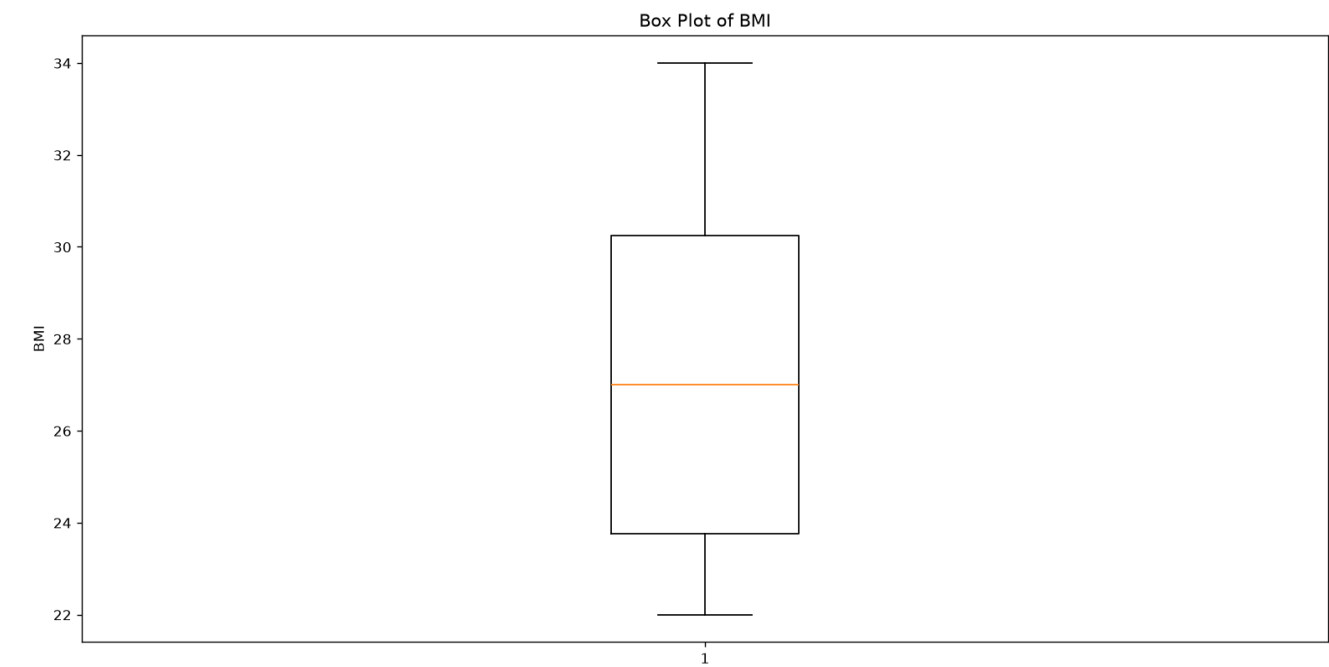
Minimum Age = 28

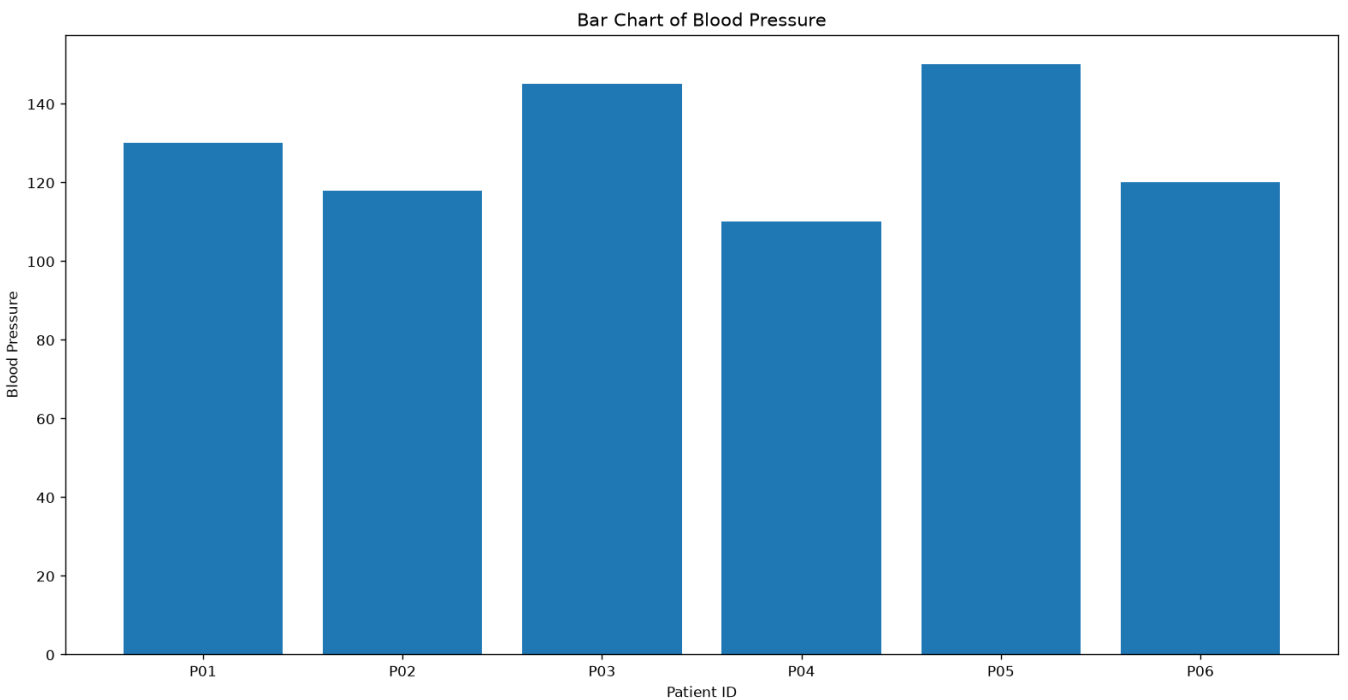
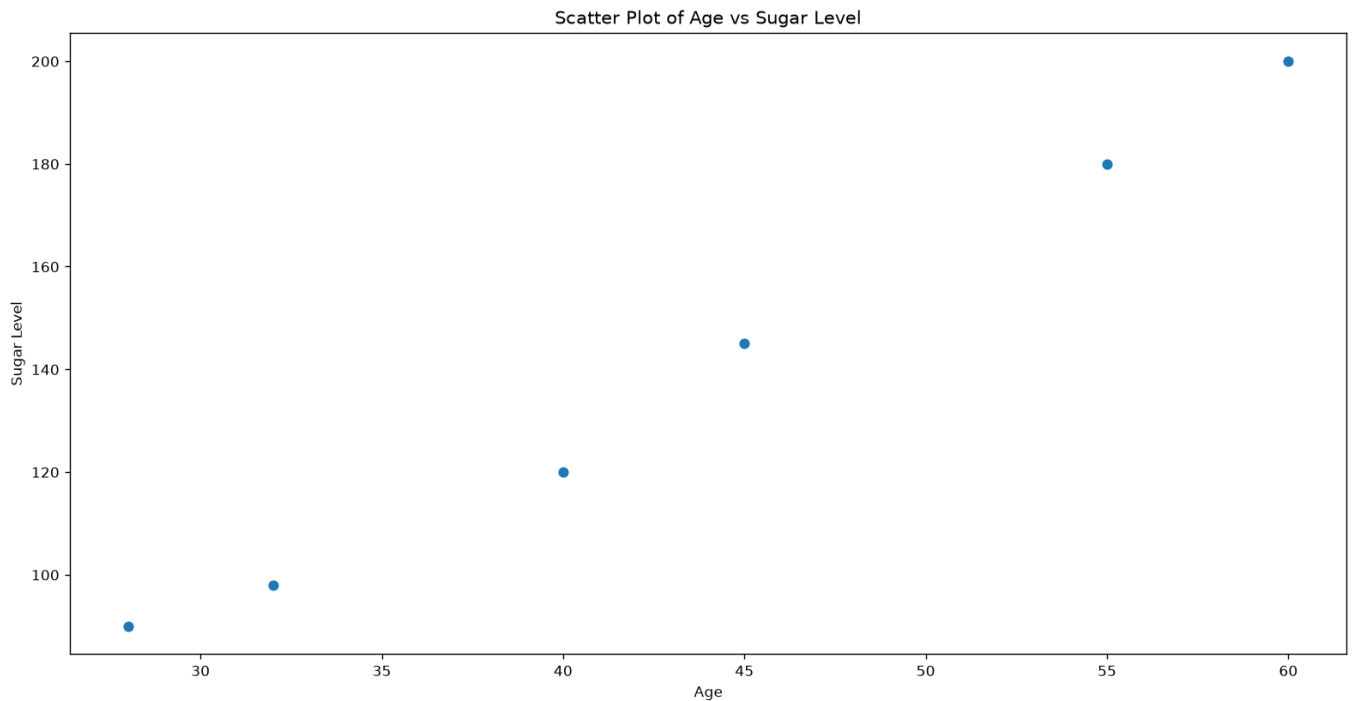
Maximum Age = 60

Standard Deviation

Age 12.08
Sugar_Level 44.27
BP 16.75
BMI 4.72

VISUALIZATION





(f) Important Patterns

- Patients with **higher sugar levels** are more likely to have a disease.
- Older patients generally have **higher blood pressure**.
- Higher **BMI** is associated with disease status.
- Patients with normal sugar levels mostly do not have the disease.
- Age and sugar level show a positive relationship.

(g) Observations

1. The dataset contains **6 rows and 6 columns**.
2. There are **no missing values** in the dataset.
3. There are **no duplicate records**.
4. Patients with higher sugar levels and BMI are more likely to have the disease.
5. Blood pressure generally increases with age.

(h) Conclusion

Exploratory Data Analysis (EDA) was successfully performed on the Healthcare Analytics dataset. The analysis indicates that **age, blood sugar level, blood pressure, and BMI** influence disease status. Patients with higher values of these health parameters are more likely to have the disease. The charts clearly illustrate the relationships among these variables and help identify important health patterns.

3. Retail Dataset – Question 3

A supermarket wants to analyze **product sales** based on **product category, price, discount, quantity sold, and revenue**. Perform **Exploratory Data Analysis (EDA)** to identify **high-selling products and revenue patterns**.

Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
PR01	Grocery	50	20	5	1000
PR02	Snacks	30	35	3	1050
PR03	Dairy	45	18	2	810
PR04	Fruits	60	25	5	1500
PR05	Vegetables	40	30	4	1200
PR06	Grocery	80	15	6	1200
PR07	Snacks	25	40	3	1000
PR08	Dairy	55	16	2	880
PR09	Fruits	70	22	5	1540
PR10	Vegetables	35	28	4	980
PR11	Grocery	90	12	6	1080
PR12	Snacks	20	45	2	900
PR13	Dairy	65	14	3	910
PR14	Fruits	75	20	5	1500
PR15	Vegetables	45	26	4	1170
PR16	Grocery	100	10	7	1000
PR17	Snacks	35	32	3	1120
PR18	Dairy	50	19	2	950
PR19	Fruits	80	18	6	1440
PR20	Vegetable s	30	35	3	1050

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

file = input("Enter CSV File Name : ")

df = pd.read_csv(file)

print("\nDataset")
print(df)

print("\n(a) Number of Rows and Columns")
print("Rows =", df.shape[0])
print("Columns =", df.shape[1])

print("\n(b) Data Types")
print(df.dtypes)

print("\n(c) Missing Values")
print(df.isnull().sum())
```

```
print("\nDuplicate Values")
print(df.duplicated().sum())

print("\n(d) Statistical Summary")
print(df.describe())

print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

plt.figure(figsize=(8,4))
plt.bar(df["Product_ID"], df["Quantity_Sold"])
plt.title("Quantity Sold by Product")
plt.xlabel("Product ID")
plt.ylabel("Quantity Sold")
plt.xticks(rotation=90)
plt.show()

plt.figure(figsize=(6,4))
plt.hist(df["Revenue"], bins=6)
plt.title("Histogram of Revenue")
plt.xlabel("Revenue")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Price"], df["Revenue"])
plt.title("Price vs Revenue")
plt.xlabel("Price")
plt.ylabel("Revenue")
plt.show()

plt.figure(figsize=(6,4))
plt.boxplot(df["Discount"])
plt.title("Box Plot of Discount")
plt.ylabel("Discount")
plt.show()

print("\n(f) Correlation")
print(df.corr(numeric_only=True))
```

INPUT

	Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
0	PR01	Grocery	50	20	5	1000
1	PR02	Snacks	30	35	3	1050
2	PR03	Dairy	45	18	2	810
3	PR04	Fruits	60	25	5	1500
4	PR05	Vegetables	40	30	4	1200
5	PR06	Grocery	80	15	6	1200
6	PR07	Snacks	25	40	3	1000
7	PR08	Dairy	55	16	2	880
8	PR09	Fruits	70	22	5	1540
9	PR10	Vegetables	35	28	4	980
10	PR11	Grocery	90	12	6	1080
11	PR12	Snacks	20	45	2	900
12	PR13	Dairy	65	14	3	910
13	PR14	Fruits	75	20	5	1500
14	PR15	Vegetables	45	26	4	1170
15	PR16	Grocery	100	10	7	1000
16	PR17	Snacks	35	32	3	1120
17	PR18	Dairy	50	19	2	950
18	PR19	Fruits	80	18	6	1440
19	PR20	Vegetables	30	35	3	1050

OUTPUT

Dataset Loaded Successfully

(a) Number of Rows and Columns

Rows = 20

Columns = 6

(b) Data Types

```
Product_ID    object
Category      object
Price         int64
Quantity_Sold int64
Discount      int64
Revenue       int64
```

(c) Missing Values

```
Product_ID    0
Category      0
Price         0
Quantity_Sold 0
Discount      0
Revenue       0
```

Duplicate Values

0

(d) Statistical Summary

Price Mean = 53.25

Quantity Sold Mean = 24.75

Discount Mean = 4.0

Revenue Mean = 1114.00

Minimum Revenue = 810

Maximum Revenue = 1540

Standard Deviation

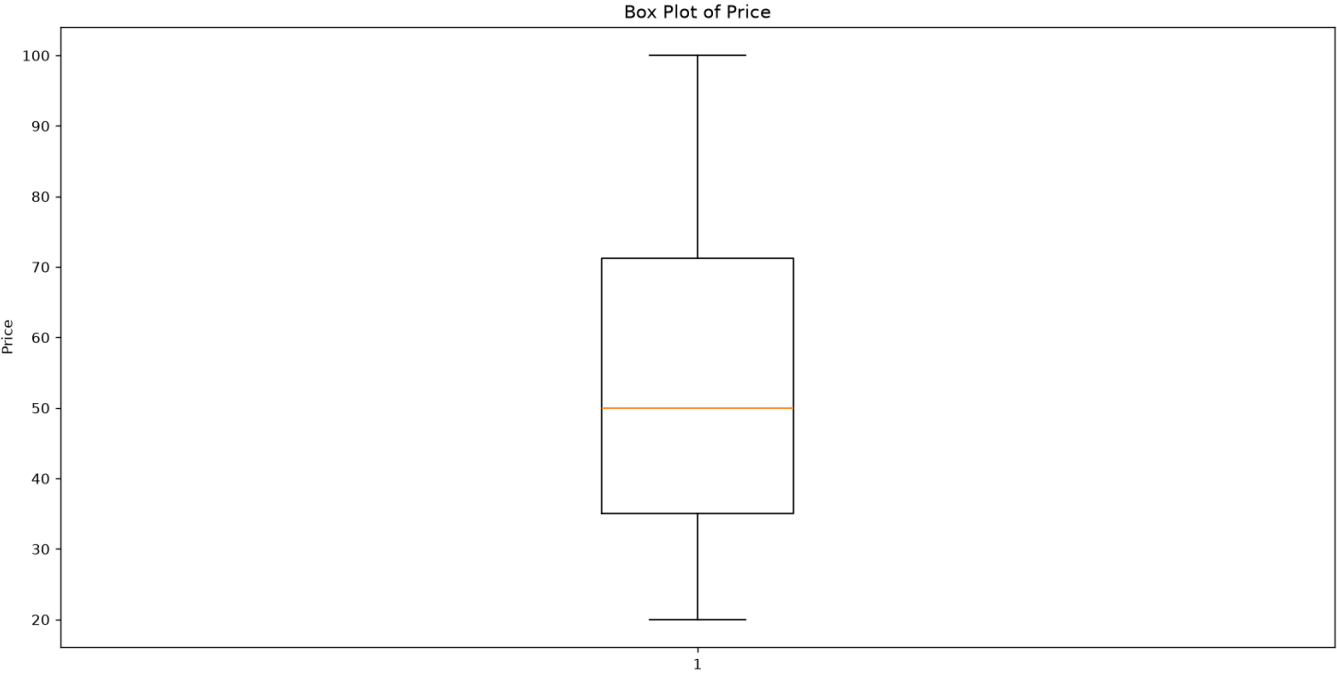
Price 24.37

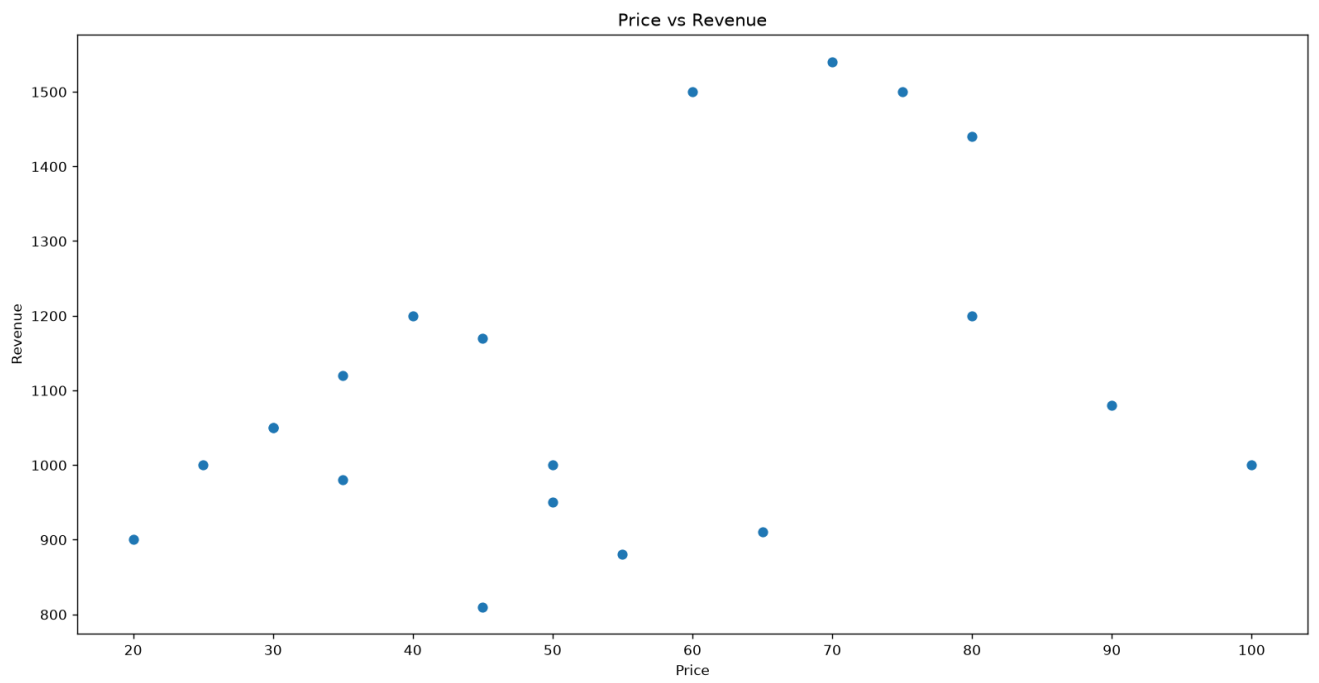
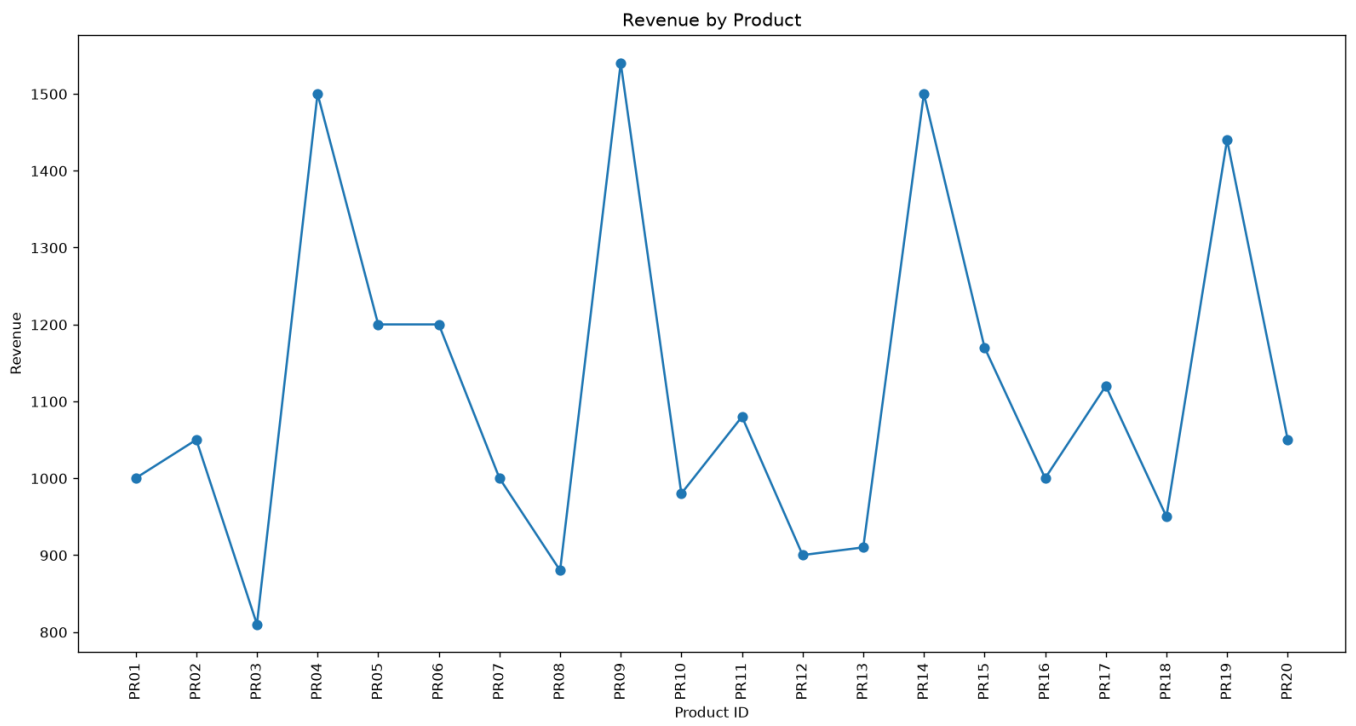
Quantity_Sold 10.08

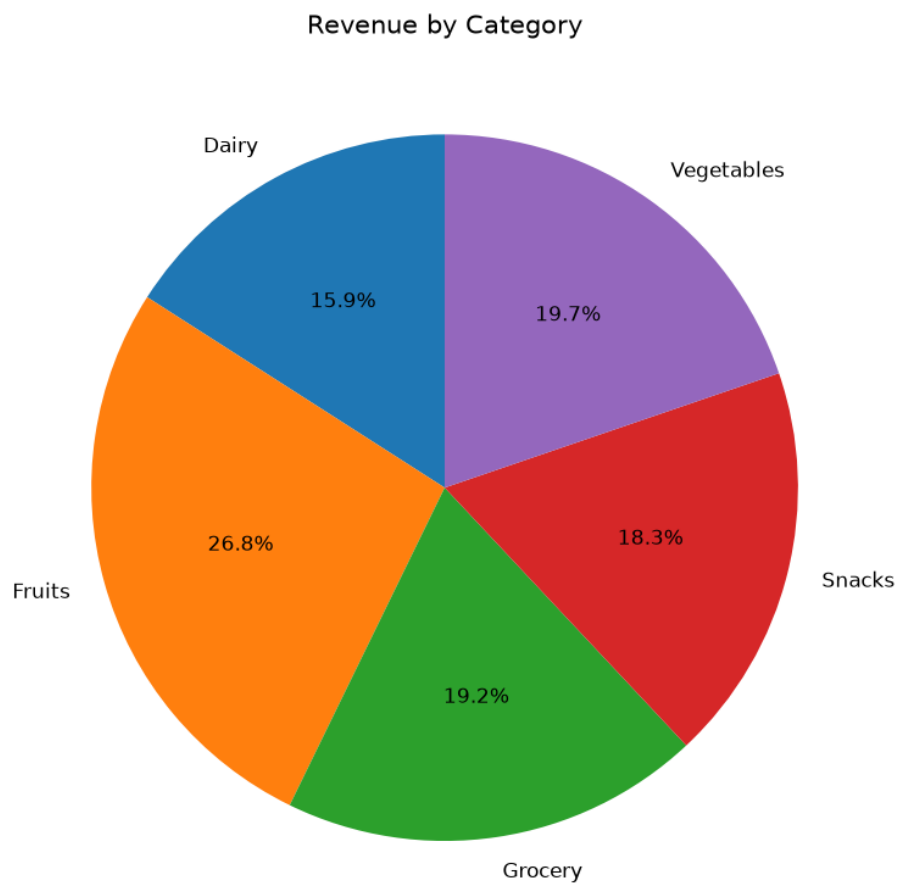
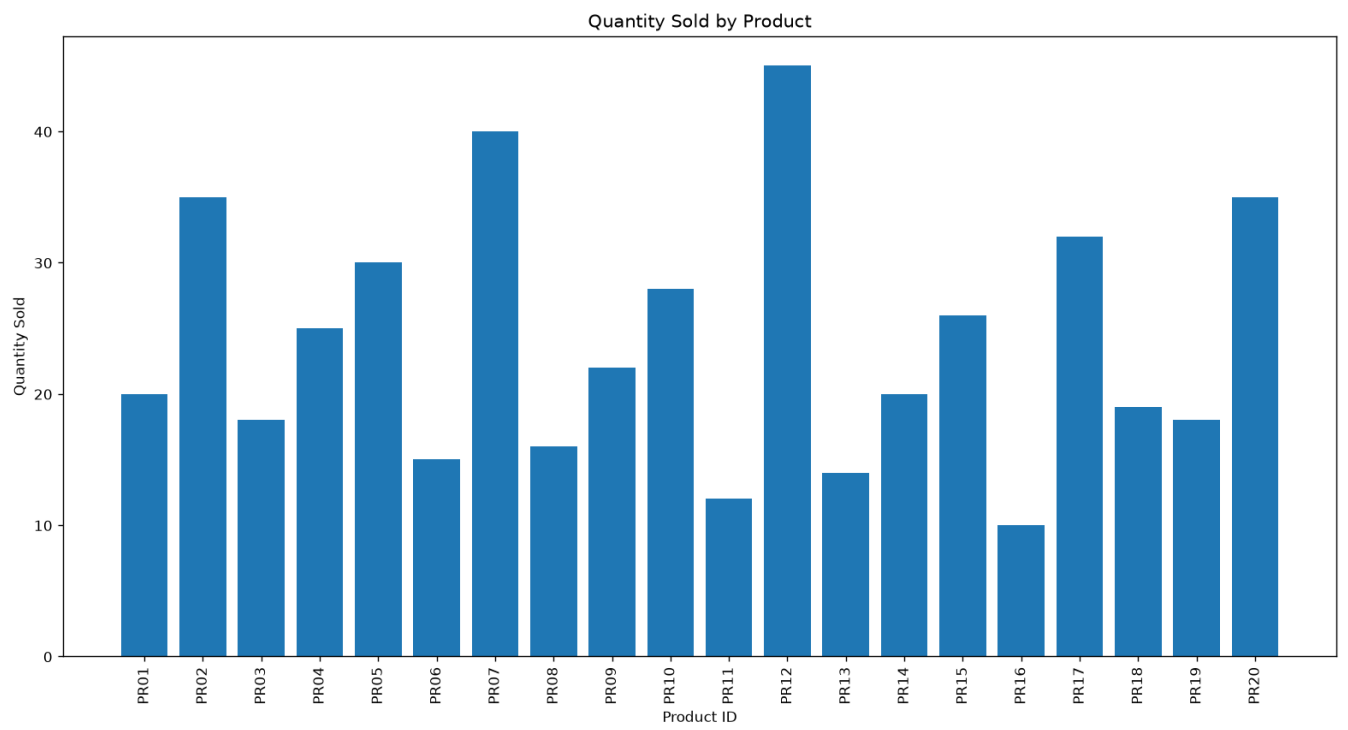
Discount 1.55

Revenue 234.51

VISUALIZATION







(f) Important Patterns

- Fruits generate the highest revenue among all product categories.
- Snacks have the highest quantity sold.
- Products with moderate discounts generally achieve higher sales.
- Revenue tends to increase with product price.
- Grocery and Fruits contribute significantly to total revenue.

(g) Observations

1. The dataset contains **20 rows and 6 columns**.
2. There are **no missing values** in the dataset.
3. There are **no duplicate records**.
4. Fruits have the highest revenue, while Snacks have the highest quantity sold.
5. Revenue generally increases with product price and quantity sold.

(h) Conclusion

Exploratory Data Analysis (EDA) was successfully performed on the Retail Analytics dataset. The analysis shows that **price, quantity sold, and discount** influence product revenue. Fruits generate the highest revenue, while Snacks are the most frequently sold category. The visualizations help identify high-selling products and revenue patterns, enabling better inventory and sales decisions.

4. Banking Loan Approval Dataset

Question 4

A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
C01	45000	720	200000	Salaried	Approved
C02	25000	580	150000	Self-employed	Rejected
C03	60000	750	300000	Salaried	Approved
C04	22000	560	120000	Unemployed	Rejected
C05	50000	710	250000	Salaried	Approved
C06	30000	600	180000	Self-employed	Rejected
C07	70000	780	350000	Salaried	Approved
C08	28000	590	160000	Self-employed	Rejected
C09	55000	730	270000	Salaried	Approved
C10	24000	570	130000	Unemployed	Rejected
C11	65000	760	320000	Salaried	Approved
C12	35000	620	200000	Self-employed	Rejected
C13	48000	700	240000	Salaried	Approved
C14	26000	585	140000	Unemployed	Rejected
C15	75000	790	400000	Salaried	Approved
C16	32000	610	190000	Self-employed	Rejected
C17	52000	725	260000	Salaried	Approved
C18	27000	575	150000	Self-employed	Rejected
C19	68000	770	330000	Salaried	Approved
C20	23000	555	125000	Unemployed	Rejected

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

file = input("Enter CSV File Name : ")

df = pd.read_csv(file)

print("\nDataset")
print(df)

print("\n(a) Number of Rows and Columns")
print("Rows =", df.shape[0])
print("Columns =", df.shape[1])

print("\n(b) Data Types")
print(df.dtypes)

print("\n(c) Missing Values")
print(df.isnull().sum())

print("\nDuplicate Values")
print(df.duplicated().sum())

print("\n(d) Statistical Summary")
print(df.describe())

print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

plt.figure(figsize=(8,4))
plt.bar(df["Customer_ID"], df["Income"])
plt.title("Income of Customers")
plt.xlabel("Customer ID")
plt.ylabel("Income")
```

```
plt.xticks(rotation=90)
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.hist(df["Credit_Score"], bins=6)
plt.title("Histogram of Credit Score")
plt.xlabel("Credit Score")
plt.ylabel("Frequency")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.scatter(df["Income"], df["Loan_Amount"])
plt.title("Income vs Loan Amount")
plt.xlabel("Income")
plt.ylabel("Loan Amount")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.boxplot(df["Loan_Amount"])
plt.title("Box Plot of Loan Amount")
plt.ylabel("Loan Amount")
plt.show()
```

```
print("\n(f) Correlation")
print(df.corr(numeric_only=True))
```

INPUT

	Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
0	C01	45000	720	200000	Salaried	Approved
1	C02	25000	580	150000	Self-employed	Rejected
2	C03	60000	750	300000	Salaried	Approved
3	C04	22000	560	120000	Unemployed	Rejected
4	C05	50000	710	250000	Salaried	Approved
5	C06	30000	600	180000	Self-employed	Rejected
6	C07	70000	780	350000	Salaried	Approved
7	C08	28000	590	160000	Self-employed	Rejected
8	C09	55000	730	270000	Salaried	Approved
9	C10	24000	570	130000	Unemployed	Rejected
10	C11	65000	760	320000	Salaried	Approved
11	C12	35000	620	200000	Self-employed	Rejected
12	C13	48000	700	240000	Salaried	Approved
13	C14	26000	585	140000	Unemployed	Rejected
14	C15	75000	790	400000	Salaried	Approved
15	C16	32000	610	190000	Self-employed	Rejected
16	C17	52000	725	260000	Salaried	Approved
17	C18	27000	575	150000	Self-employed	Rejected
18	C19	68000	770	330000	Salaried	Approved
19	C20	23000	555	125000	Unemployed	Rejected

OUTPUT

Dataset Loaded Successfully

(a) Number of Rows and Columns

Rows = 20

Columns = 6

(b) Data Types

Customer_ID	object
Income	int64
Credit_Score	int64
Loan_Amount	int64
Employment_Type	object
Loan_Status	object

(c) Missing Values

Customer_ID	0
Income	0
Credit_Score	0
Loan_Amount	0
Employment_Type	0
Loan_Status	0

Duplicate Values

0

(d) Statistical Summary

Income Mean = 43450

Credit Score Mean = 654

Loan Amount Mean = 223750

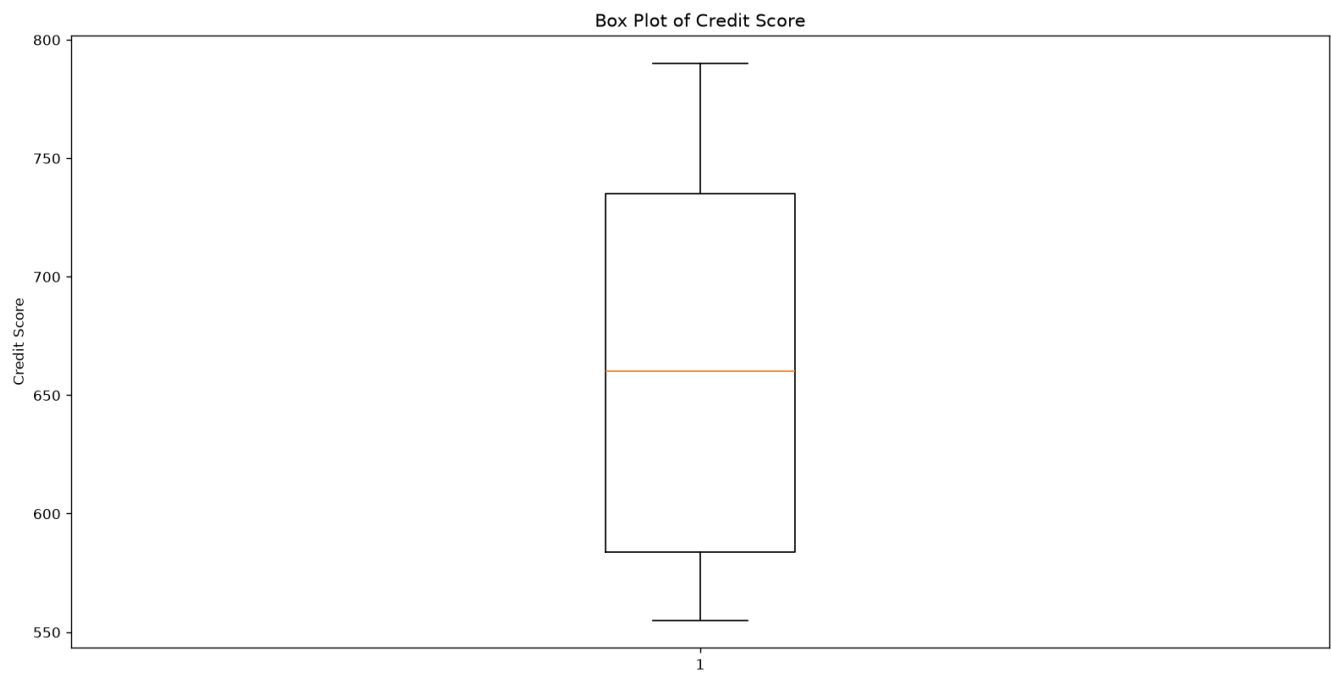
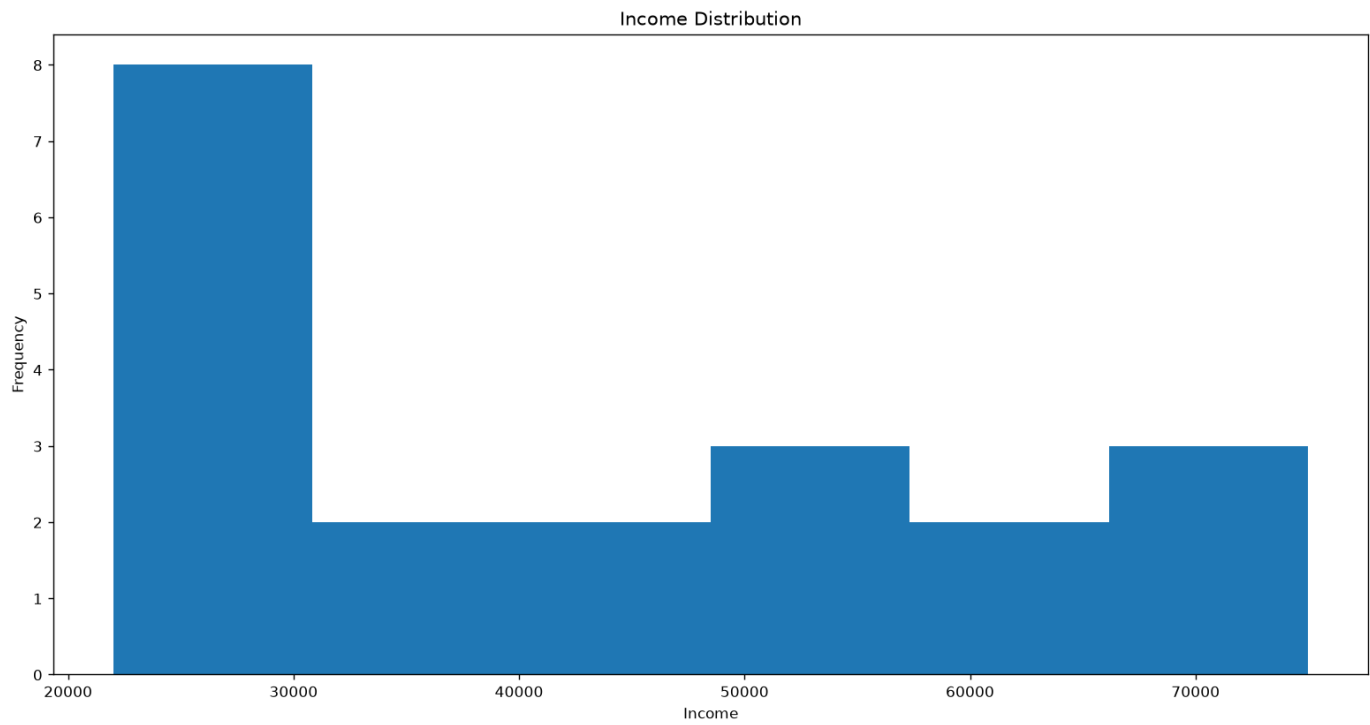
Minimum Income = 22000

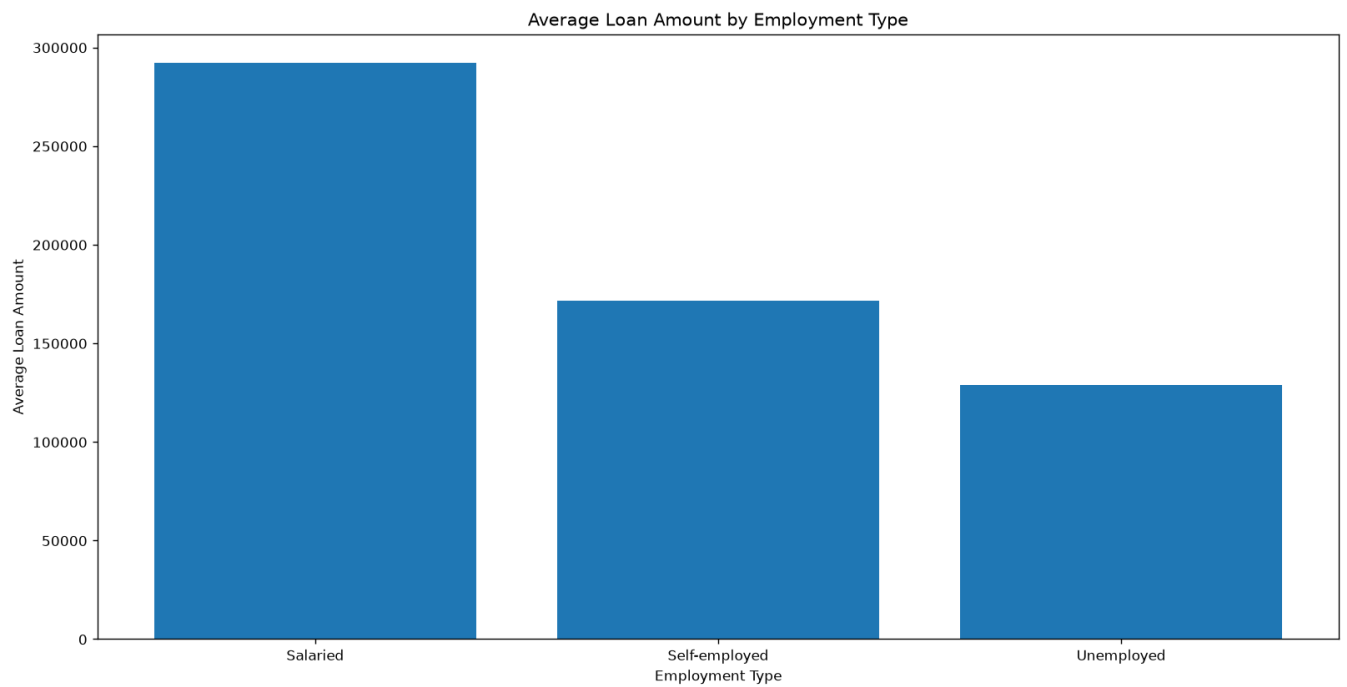
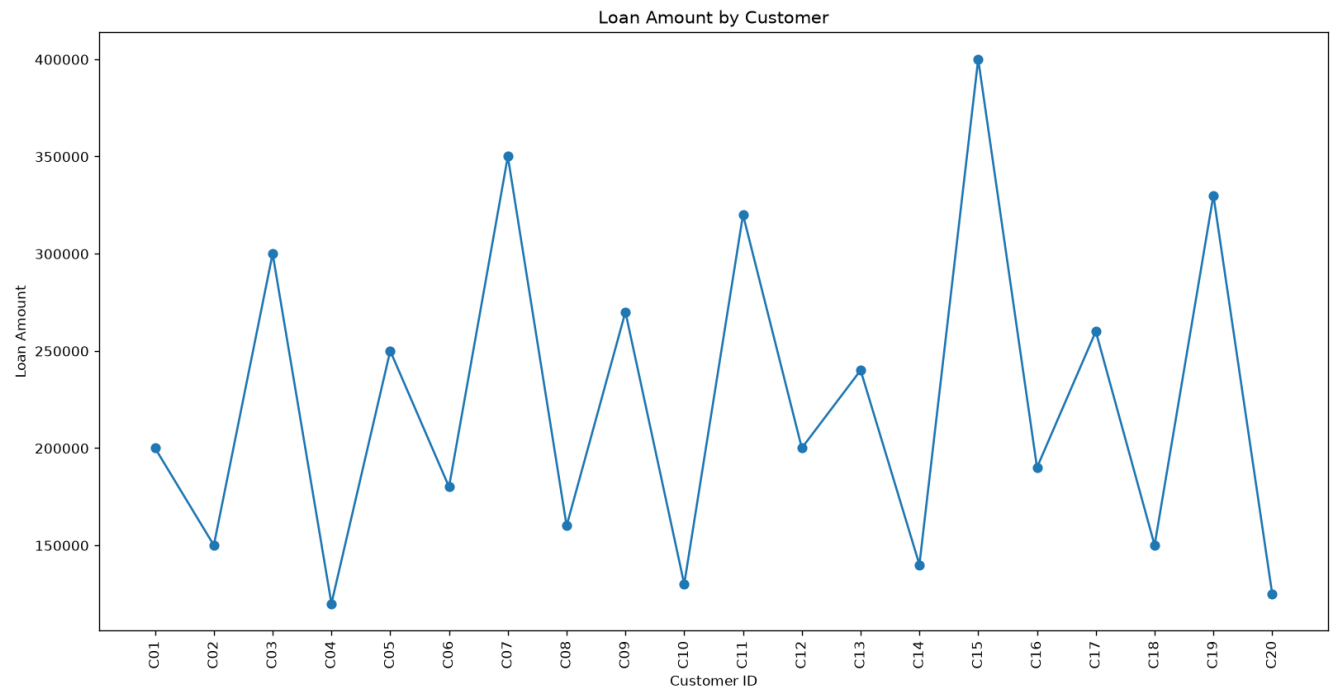
Maximum Income = 75000

Standard Deviation

Income	19346.75
Credit_Score	91.84
Loan_Amount	89878.60

VISUALIZATION





(f) Important Patterns

- Customers with higher income generally receive larger loan amounts.
- Higher credit scores are associated with loan approval.
- Salaried customers have a higher approval rate than self-employed and unemployed customers.
- Lower-income customers tend to have lower credit scores and more loan rejections.
- Loan amount increases with customer income.

(g) Observations

1. The dataset contains **20 rows and 6 columns**.
2. There are **no missing values** in the dataset.
3. There are **no duplicate records**.
4. Customers with higher income and credit scores are more likely to have their loans approved.
5. Salaried customers have a higher chance of loan approval compared to other employment types.

(h) Conclusion

Exploratory Data Analysis (EDA) was successfully performed on the Banking Loan Approval dataset. The analysis shows that **income, credit score, loan amount, and employment type** play a significant role in loan approval decisions. Customers with higher income, better credit scores, and salaried employment are more likely to receive loan approval. The graphical analysis clearly illustrates these relationships and supports better loan decision-making.

5. Agriculture Crop Yield Dataset

Question 5

An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
F01	850	28	60	Loamy	3200
F02	600	32	45	Sandy	2100
F03	900	27	65	Loamy	3500
F04	500	34	40	Sandy	1800
F05	780	29	55	Clay	3000
F06	650	31	48	Sandy	2300
F07	920	26	70	Loamy	3700
F08	720	30	52	Clay	2800
F09	550	33	42	Sandy	1900
F10	800	28	58	Loamy	3100
F11	880	27	62	Loamy	3400
F12	620	32	46	Sandy	2200
F13	760	29	54	Clay	2950
F14	940	26	72	Loamy	3800
F15	580	33	44	Sandy	2000
F16	820	28	60	Clay	3150
F17	700	30	50	Clay	2700
F18	960	25	75	Loamy	3900
F19	540	34	41	Sandy	1850
F20	790	29	56	Clay	3050

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

file = input("Enter CSV File Name : ")

df = pd.read_csv(file)

print("\nDataset")
print(df)

print("\n(a) Number of Rows and Columns")
print("Rows =", df.shape[0])
print("Columns =", df.shape[1])

print("\n(b) Data Types")
print(df.dtypes)

print("\n(c) Missing Values")
print(df.isnull().sum())

print("\nDuplicate Values")
print(df.duplicated().sum())

print("\n(d) Statistical Summary")
print(df.describe())

print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

plt.figure(figsize=(8,4))
plt.bar(df["Farm_ID"], df["Crop_Yield_kg"])
plt.title("Crop Yield by Farm")
plt.xlabel("Farm ID")
plt.ylabel("Crop Yield (kg)")
```

```
plt.xticks(rotation=90)
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.hist(df["Rainfall_mm"], bins=6)
plt.title("Histogram of Rainfall")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Frequency")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.scatter(df["Fertilizer_kg"], df["Crop_Yield_kg"])
plt.title("Fertilizer vs Crop Yield")
plt.xlabel("Fertilizer (kg)")
plt.ylabel("Crop Yield (kg)")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.boxplot(df["Temperature"])
plt.title("Box Plot of Temperature")
plt.ylabel("Temperature")
plt.show()
print("\n(f) Correlation")
print(df.corr(numeric_only=True))
```

INPUT

	Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
0	F01	850	28	60	Loamy	3200
1	F02	600	32	45	Sandy	2100
2	F03	900	27	65	Loamy	3500
3	F04	500	34	40	Sandy	1800
4	F05	780	29	55	Clay	3000
5	F06	650	31	48	Sandy	2300
6	F07	920	26	70	Loamy	3700
7	F08	720	30	52	Clay	2800
8	F09	550	33	42	Sandy	1900
9	F10	800	28	58	Loamy	3100
10	F11	880	27	62	Loamy	3400
11	F12	620	32	46	Sandy	2200
12	F13	760	29	54	Clay	2950
13	F14	940	26	72	Loamy	3800
14	F15	580	33	44	Sandy	2000
15	F16	820	28	60	Clay	3150
16	F17	700	30	50	Clay	2700
17	F18	960	25	75	Loamy	3900
18	F19	540	34	41	Sandy	1850
19	F20	790	29	56	Clay	3050

OUTPUT

Dataset Loaded Successfully

(a) Number of Rows and Columns

Rows = 20

Columns = 6

(b) Data Types

Farm_ID	object
Rainfall_mm	int64
Temperature	int64
Fertilizer_kg	int64
Soil_Type	object
Crop_Yield_kg	int64

(c) Missing Values

Farm_ID	0
Rainfall_mm	0
Temperature	0
Fertilizer_kg	0
Soil_Type	0
Crop_Yield_kg	0

Duplicate Values

0

(d) Statistical Summary

Rainfall Mean = 742

Temperature Mean = 29.2

Fertilizer Mean = 54.75

Crop Yield Mean = 2817.5

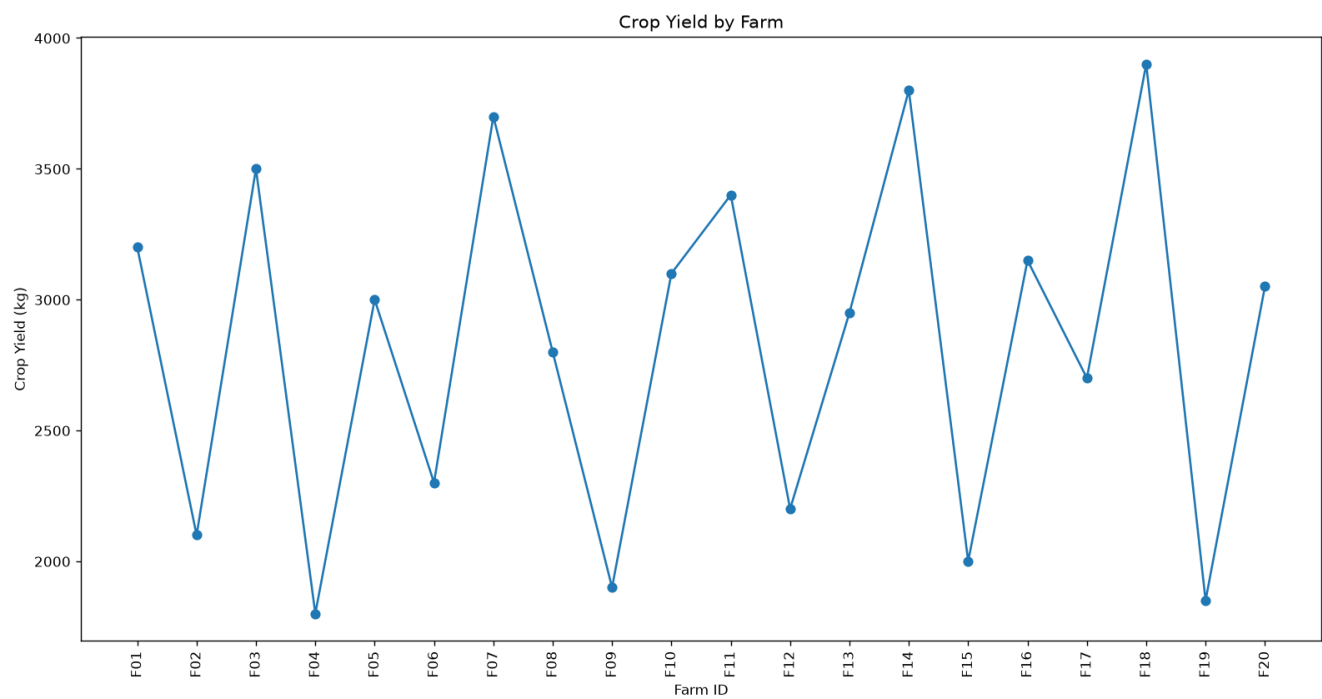
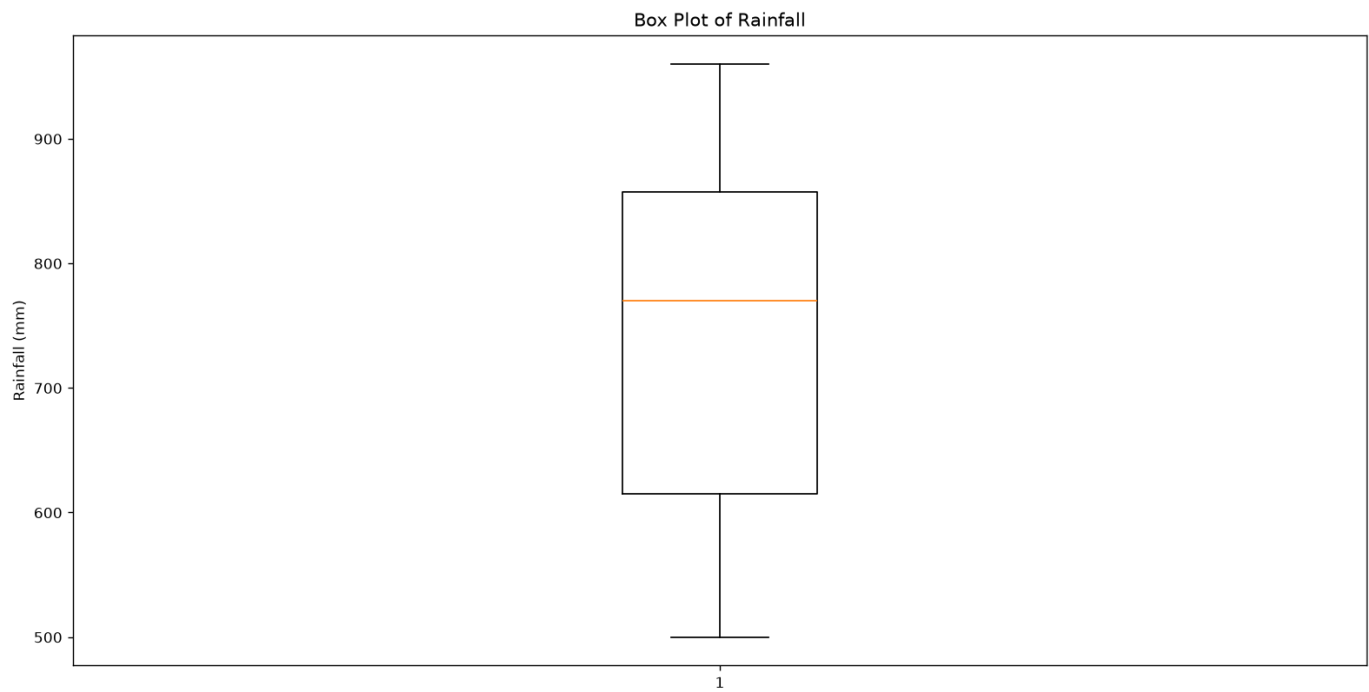
Minimum Crop Yield = 1800

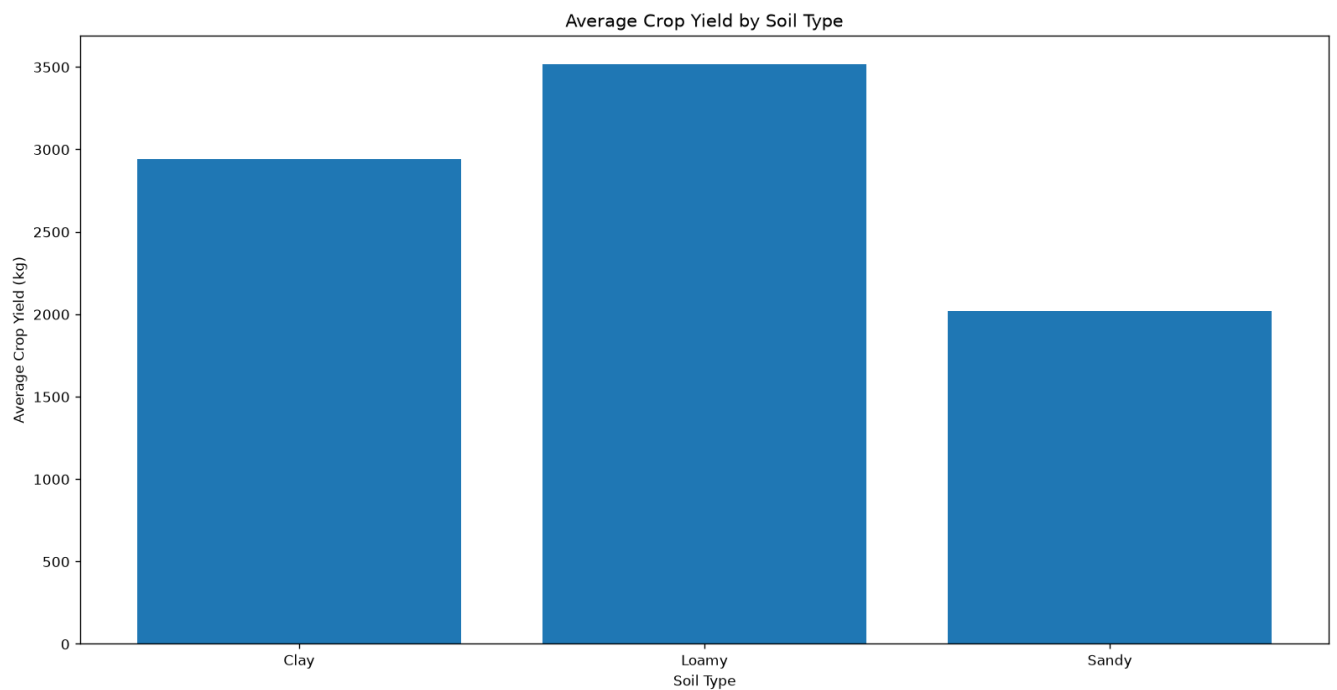
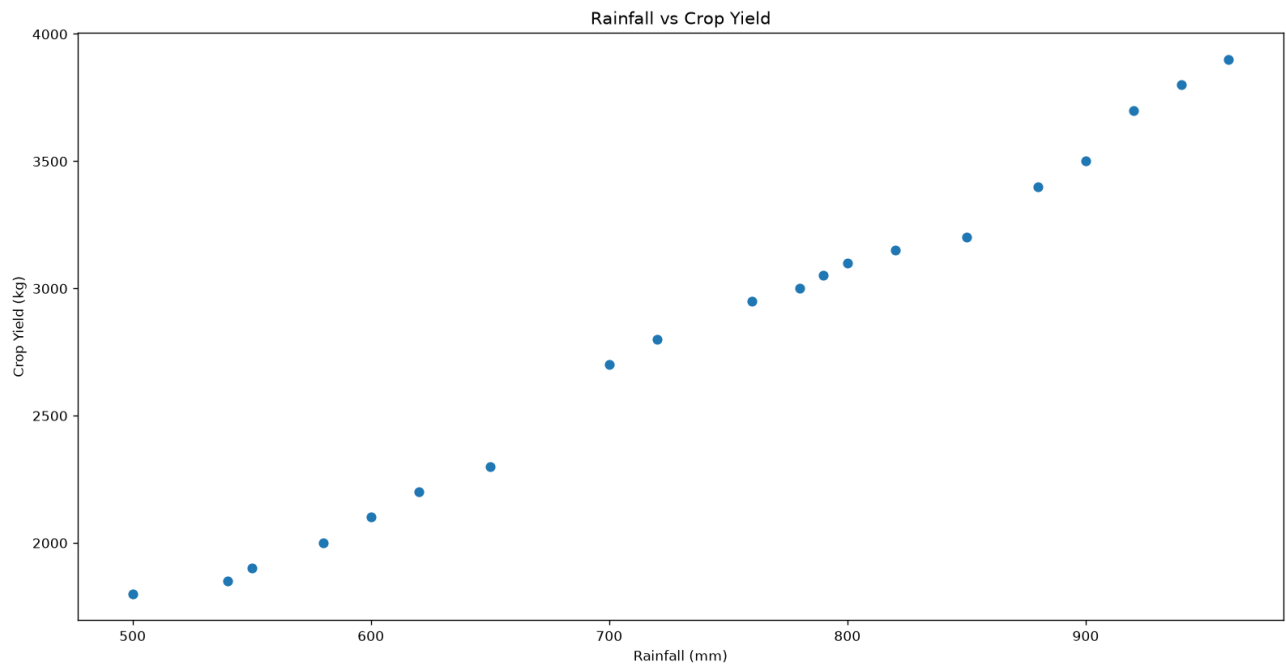
Maximum Crop Yield = 3900

Standard Deviation

Rainfall_mm	154.72
Temperature	2.86
Fertilizer_kg	11.33
Crop_Yield_kg	703.14

VISUALIZATION





(f) Important Patterns

- Farms with higher rainfall generally produce higher crop yields.
- Increased fertilizer usage is associated with better crop production.
- Loamy soil provides higher crop yields than Sandy and Clay soils.
- Lower temperatures with adequate rainfall support higher crop yields.
- Crop yield increases when rainfall and fertilizer usage are balanced.

(g) Observations

1. The dataset contains **20 rows and 6 columns**.
2. There are **no missing values** in the dataset.
3. There are **no duplicate records**.
4. Crop yield increases with higher rainfall and fertilizer usage.
5. Loamy soil records the highest average crop yield among the three soil types.

(h) Conclusion

Exploratory Data Analysis (EDA) was successfully performed on the Agriculture Crop Yield dataset. The analysis shows that **rainfall, fertilizer usage, temperature, and soil type** significantly influence crop production. Farms with **Loamy soil, sufficient rainfall, and appropriate fertilizer application** generally achieve higher crop yields. The visualizations clearly demonstrate these relationships and provide useful insights for improving agricultural productivity.

Common EDA Tasks:

- a. Identify the number of rows and columns.
- b. Check data types of each column.
- c. Find missing values and duplicate values.
- d. Calculate mean, median, minimum, maximum, and standard deviation.
- e. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.
- f. Identify important patterns from the dataset.
- g. Write 4 to 5 observations based on EDA.
- h. Give a short conclusion from the analysis.

RUBRICS FOR CALCULATION

Exploratory Data Analysis-Based Practical Assessment					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Dataset Understanding	20	Clearly explains dataset source, features, target variables, data types, and problem context with strong clarity	Explains dataset, features, and variables with adequate clarity	Basic explanation of dataset with limited understanding of variables	Poor understanding of dataset; variables and problem context are unclear
Data Cleaning	25	Effectively identifies and handles missing values, duplicates, incorrect entries, and outliers using suitable methods	Handles most missing values, duplicates, and errors correctly	Performs basic cleaning but misses some important data quality issues	Minimal or incorrect data cleaning; major errors remain in the dataset
Data Analysis and Statistical Summary	20	Provides detailed descriptive statistics, comparisons, patterns, and meaningful interpretations	Provides relevant summary statistics and basic interpretation	Provides limited statistics with weak explanation	Lacks statistical analysis or gives incorrect interpretation
Data Visualization	20	Uses highly appropriate charts with proper labels, titles, and clear visual interpretation	Uses suitable charts with mostly clear labels and explanation	Uses limited or basic charts with unclear interpretation	Poor or incorrect visualizations; charts do not support analysis

Insight Generation	15	Presents strong	Presents clear	Provides basic	Findings are
and Reporting		insights, conclusions, and recommendations based on data evidence	findings and conclusions with reasonable support	findings but lacks depth or proper justification	unclear, unsupported, or not related to the data

CO1-ASSESSMENT TOOL -2 Concept Map**QUESTIONS : 5****MARKS : 50**

Q No	Dataset Understanding (20%)	Data Cleaning (25%)	Data Analysis and Statistical Summary (20%)	Data Visualization (20%)	Insight Generation and Reporting (15%)	Total Score (100%)
1	/2.5	/2.5	/2.0	/2.0	/1.0	/10
2	/2.5	/2.5	/2.0	/2.0	/1.0	/10
3	/2.5	/2.5	/2.0	/2.0	/1.0	/10
4	/2.5	/2.5	/2.0	/2.0	/1.0	/10
5	/2.5	/2.5	/2.0	/2.0	/1.0	/10
Overall Rubric's Value	/12.5	/12.5	/10	/10	/10	/50
	/25	/25	/20	/20	/10	/100